15.10 A Vertical-Cell-Transistor-Based 4F² DRAM with Cell-on-Peripheral Architecture Using Wafer-to-Wafer Hybrid Copper Bonding

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Abstract

The paper presents a 16Gb SDRAM with a vertical-channel-transistor-based 4F² cell and cell-on-peripheral (COP) architecture. Wafer-to-wafer hybrid copper bonding is applied to COP, which increases the total die count per wafer by more than 20% with the same design rules. Fabricated in a 10nm-class DRAM process, the test chip performs read and write operations successfully, opening the possibility for further DRAM-technology scaling.

The scalability of dynamic random-access memory (DRAM), which integrates capacitors and transistors within a 2-dimensional environment, is facing physical limitations. On the other hand, industry trends consistently demand higher density and bandwidth at a reduced energy-per-bit. To address these challenges, new DRAM architectures are being actively investigated [1,2,3]. In particular, the vertical channel transistor (VCT), in which the source and drain are separated along different planes in a vertical configuration, has been investigated for the DRAM cell [1]. However, the device is fabricated in a monolithic process, hence it is difficult to form well-defined buried bit-line (BL); VCT also suffer from the floating-body effect [2]. In addition, recent research has proposed a 4F² DRAM cell using VCT and wafer-bonding techniques for next-generation DRAM to overcome monolithic-integration limitations [3]. However, the vertical interface to form inter-wafer contacts requires additional area, since each VCT's BL and WL must be connect to the BL sense amplifier (BLSA), which detects the data polarity stored in a DRAM cell, and a sub-WL driver (SWD) to drive the WL, following the wafer-to-wafer bonding process. Meanwhile, as Moore's law slows down the application of heterogeneous 3D integrated circuits is being developed with hybrid copper bonding (HCB). The wafer-to-wafer HCB process has already been commercialized for CMOS image sensor and NAND Flash products [4,5]. HCB is a good alternative for VCT DRAM, which requires wafer bonding techniques.

This paper proposes a new DRAM architecture to replace conventional DRAM for memory-process scaling extension. With sub-10nm VCT, wafer-to-wafer HCB achieves a gross die gain of over 20%, compared to a conventional architecture, by separating the cell and the peripheral circuits (PERI) area, via a cell-over-PERI (COP) architecture. With the adoption of a vertical cell structure, the parasitic capacitance of the BL is reduced. In addition, improved design flexibility for the BLSA and SWD allows for the number of cells on each WL and BL to be freely adjusted, resulting in a 10% and 30% reduction in bank active-precharge current I_{DD0} and burst refresh current I_{DD5}, without compromising additional area.

Device scaling inherently reduces the capacitance of the charge-storage capacitor (C_S), making reliable data retention challenging. Furthermore, as the space between adjacent cells diminishes, the row-hammer (RH) effect, which causes charge loss when adjacent WLs are frequently accessed, becomes more pronounced. Figure 15.10.1 (top-left) shows a VCT cell structure to address challenges faced in DRAM cell shrinkage. The vertical channel mitigates short-channel effects, as the gate length is vertical height of the channel, without compromising area. Moreover, its bulk-less architecture prevents charge loss through the body when adjacent cells are operating, unlike conventional structure that suffer from RH. Figure 15.10.1(bottom-right) shows a top view of the simplified 4 cells. In the VCT architecture, spatial decoupling of the storage node from the BL leads to a significant reduction in BL parasitic capacitance (C_{BL}). Figure 15.10.1(bottom-left) shows an overview of the VCT DRAM architecture, which consists of cell and PERI wafers, indicating the wafer bonding process. Figure 15.10.1 (top-right) illustrates a comparison of the overall vertical structure, highlighting the COP architecture connected via HCB. The HCB process, which fabricates cell and PERI wafers independently and bonds the two wafers via a Cu pad, can connect every BL and WL to the corresponding BLSA and SWD.

The most critical change, due the COP architecture, is the placement of the DRAM cells and PERI on different wafers, followed by the process required to bond two wafers into a single device. The major factor of COP architecture strongly depends on how effectively the DRAM core circuits, represented by BLSA and SWD, are rearranged when adopting the HCB process. Figure 15.10.2 shows a variety of arrangements using the COP architecture. Figure 15.10.2(top-left) shows the conventional architecture where the BLSA and SWD are positioned adjacent to the cell array, aligned respectively with the directions of the BL and WL. However, with a COP architecture, the core circuits can be placed above the cell array since cell and PERI are located on different wafers. Figure 15.10.2(top-right) shows two COP architectures with BLSAs and SWDs respectively overlapped on the cell array. Since the BLSA must detect small signal differences, the strict requirements for sensitivity and reliability pose significant challenges to continued scaling [6,7]. Moreover, typically BLSAs occupy area that is more than double of the SWD, thus it is advantageous to overlay the BLSAs on the cell array. Meanwhile, the SWDs also must fit within a restricted area defined by the fixed cell array length (or width) to directly connect WLs for minimum routing length, which makes it prone to short-channel effects and reliability issues due to reducing the cell size results in a shorter transistor's length. However, it is difficult to extend SWD's area as it directly affects chip size. Therefore, it is optimal to arrange both the BLSAs and SWD circuits to maximize the utilization of the cell array area without consuming additional space. A structural approach is proposed to address the spatial limitations of the core circuitry. As shown Fig. 15.10.2(bottom-left), the windmill and checkerboard configurations arrange the orthogonal BLSAs and SWDs to avoid overlaps, whereas the zigzag placement of the core circuits results in complex connectivity with PERI. In addition, it is not possible for both the BLSAs and SWDs to overlap at the outermost edge of cell array, hence there is a significant loss in chip area. Alternatively, the sandwich structure, shown in Fig. 15.10.2(bottom-right), independently isolates the BLSAs and SWDs area, resulting in a more straightforward routing flow. Furthermore, SWDs can fully overlay with the cell array even at the edge region, and the BLSAs can overlap by up to half. Therefore, the prototype adopts the sandwich architecture.

Thanks to the COP architecture, the size of the core transistors has a negligible impact on the entire chip size, which allows for high flexibility in applying the number of cells per BL and cells per WL. Traditionally, many cells share one BL, decreasing the required number of BLSAs required; thus, improving cost effectiveness. The greater the number of cells connected to a single BL leads to a degradation of the BLSA's read/write margin, which is a major obstacle to DRAM scaling. Figure 15.10.3(top) shows the trade-off relationship between chip size and BLSA performance for different cells/BL. The conventional approach has a strong negative correlation, while the COP architecture exhibits a relationship that significantly reduced correlation compared to conventional one. Meanwhile, a smaller cell/BL ratio, without compromising additional area for BLSAs, can decrease I_{DD0} and greatly diminishes I_{DD5}, consumed by the DRAM cell's data retention. Furthermore, core circuits represented by BLSA and SWD are designed to occupy an area equivalent to the cell region. As shown in Fig. 15.10.3 (bottom), the approximately 4:1 ratio between cell and core areas enables the expansion of the core transistor's size resulting in 2× improved sensitivity and about 10 times improved on-off current ratio of BLSA's transistors.

Although the COP architecture, using HCB, offers numerous advantages, it inherently possesses certain drawbacks. The interconnection between cells and core circuits requires additional wires resulting in increasing parasitic capacitance and resistance. Figure 15.10.4(left) shows the representative additional wires of the COP architectures. In contrast to the windmill configuration, the sandwich architecture exhibits a longer WL route, while shortening the BL additional wiring to less than half of the windmill's BLSA area. In general, BLSA performance is more critical compared to SWD, thus the sandwich topology is appropriate. In addition, the face-to-face bonding of wafers requires electrical connections passing through every layer of the top and bottom wafers as shown in Fig. 15.10.4(right). It is estimated that amount of the layer consumption accounts for about 20% to 30% of each metal layer on the bottom wafer containing the PERI transistors depending on its design rules. To address this issue, circuit-level efforts are required to reduce the number of interconnects needed to access of cell array. Figure 15.10.5 illustrates circuit techniques used to minimize the number of routes required for cell array access. First, to reduce the input signals of the SWD for WL driving, the number of input signals can be minimized by 75%, as shown in Fig 15.10.5(top). While additional transistors are required for the NOR-type SWD topology, the 4× increase in SWD area within the COP architecture is sufficient to accommodate the placement of additional transistors. Next, as shown Fig. 15.10.5(bottom), the technique achieves a 50% reduction in the number of column address select lines (CSL) by using 2:1-column multiplexers located within the enlarged core area: using the LSB of the column address to choose between two CSLs. However, a series-connected 2:1 multiplexers increase the resistance of the corresponding signal path, leading to degradation of read/write performance. Nonetheless, these technologies are essential for obtaining a sufficient back-end-of-line (BEOL) budget required for HCB.

The prototype DRAM is fabricated in a sub-10nm DRAM process. The array size is 1296cell/BL×1056cell/WL, corresponding to a typical array configuration for commercial DRAM products, see Fig. 15.10.7. A performance variations when data is written to the cell, at different temperatures, are shown in Fig 15.10.6(top). As for conventional DRAM cells, the number of failing bits significantly increases at -25° compared to 95°C, however, this failure level is manageable and repairable. The measured failed bit count per chip versus

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data retention time is shown Fig. 15.10.6(bottom). In general, 1s tend to be more vulnerable to off-state leakage, compared to 0s in conventional DRAMs; whereas, for VCT 0s are more susceptible due to the floating-body effect. Nevertheless, the overall quantity of failed bits versus retention time shows comparable results to that of a conventional DRAM. These measurement results prove that VCT- based DRAM cells with sub-10nm scaling are competitive in comparison to conventional DRAM; indicating the possibility of replacing conventional DRAM for memory process scaling extension.

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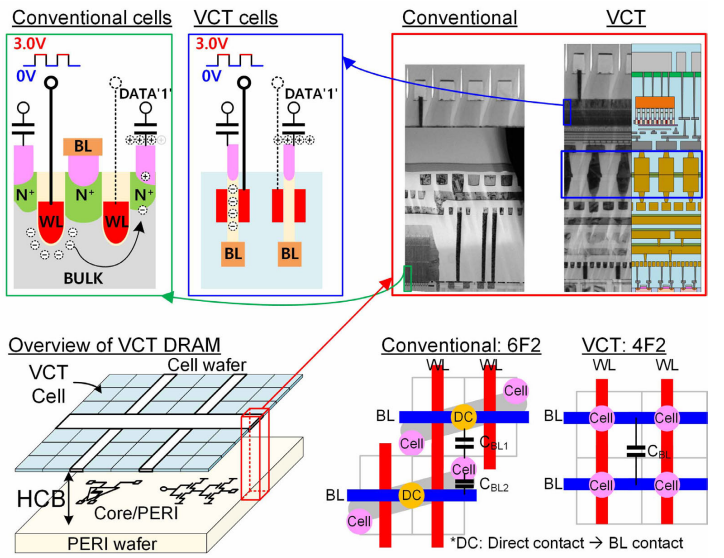

Figure 15.10.1: Comparison between a conventional and VCT architecture: cross section of unit cells (top-left), full integration (top-right), and top view of unit cells (bottom-right).

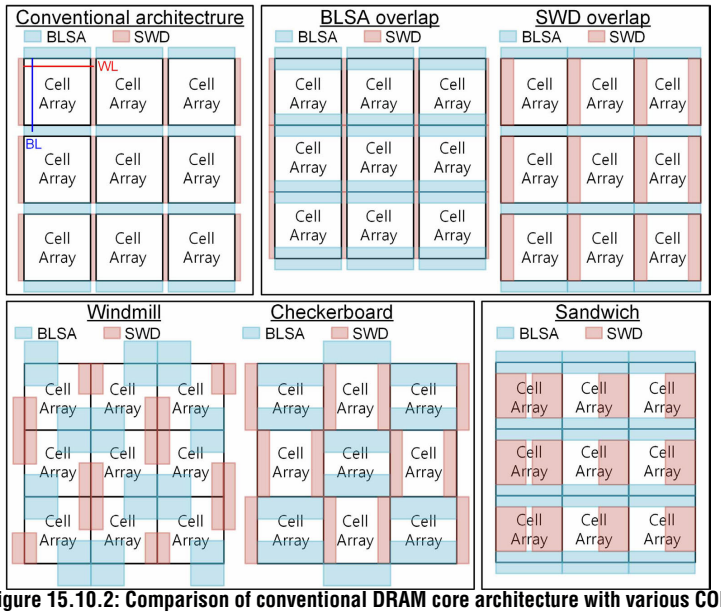

Figure 15.10.2: Comparison of conventional DRAM core architecture with various COP architectures.

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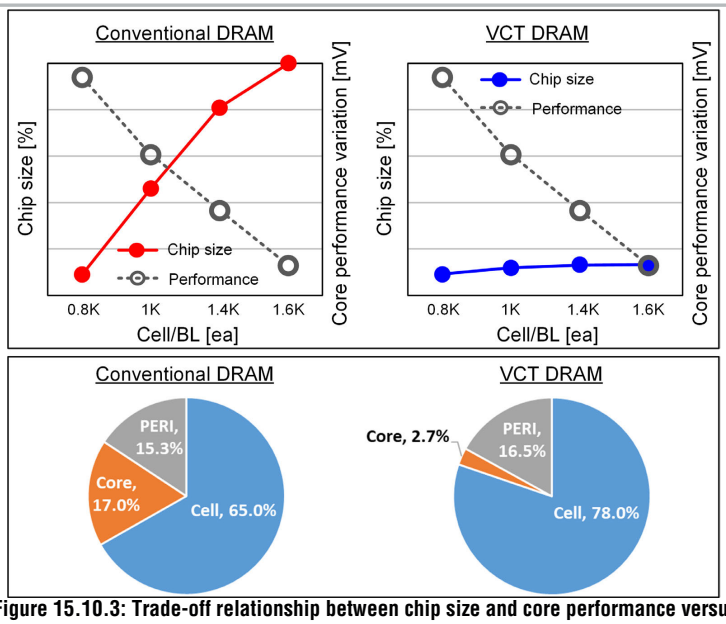


Figure 15.10.3: Trade-off relationship between chip size and core performance versus the different cell/BL (top) and comparison of the chip size portion (bottom).

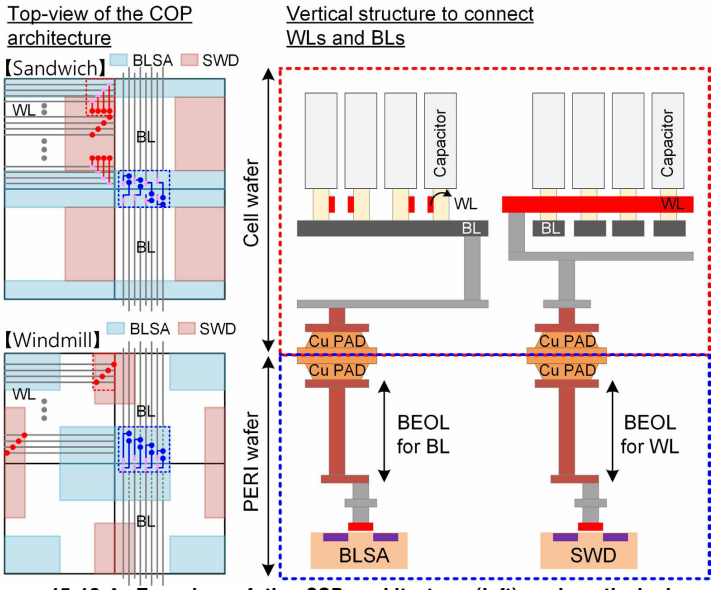


Figure 15.10.4: Top view of the COP architecture (left) and vertical view of interconnection between top and bottom wafer using HCB (right).

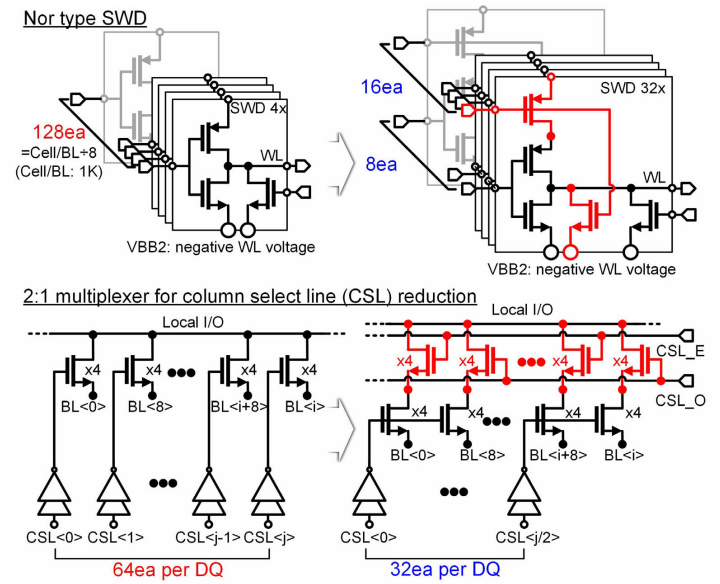


Figure 15.10.5: Circuit technology to reduce interconnection on array: NOR type SWD scheme (top) and 2:1 multiplexer for column-select reduction (bottom).

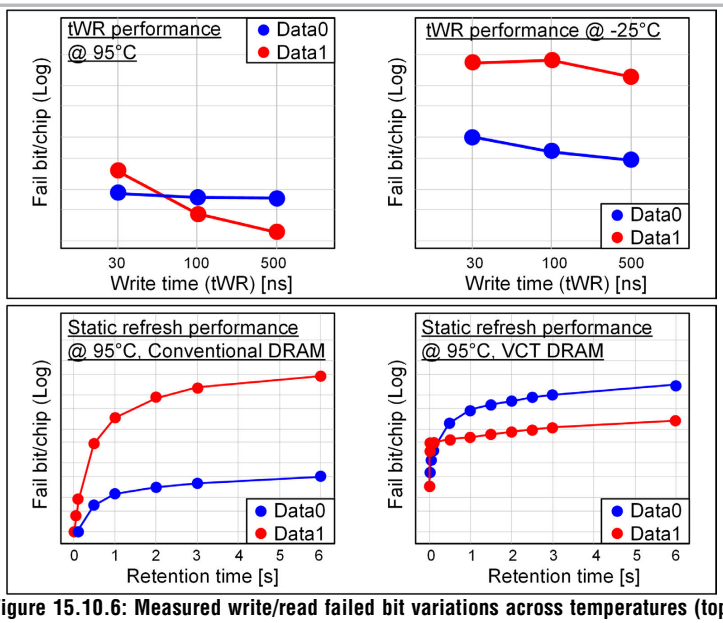
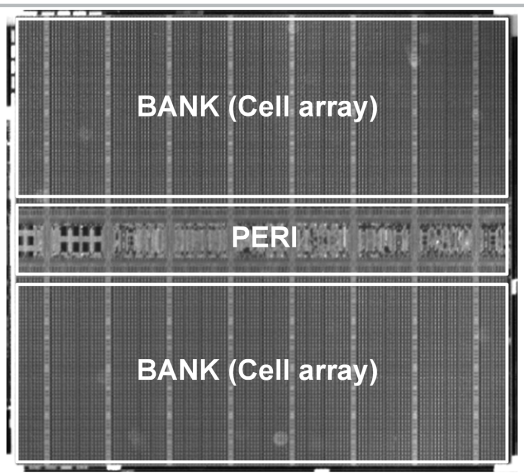


Figure 15.10.6: Measured write/read failed bit variations across temperatures (top) and static refresh failed bits for conventional and VCT DRAM (bottom).



Density	16 Gb, 32 Bank
Cell/BL per unit array	1296 Cell/BL
Cell/WL per unit array	1024 Cell/WL

Figure 15.10.7: Chip micro photograph and array configuration summary.

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